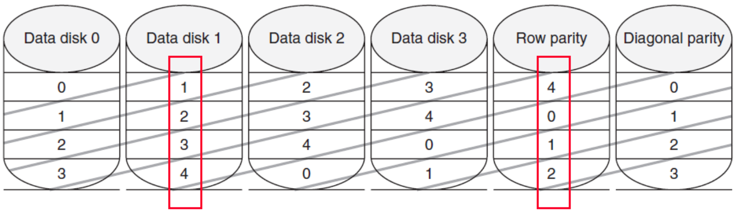
1102 Final (Exam Time: 2 hours)

1. (15%) Assume a write invalidate cache coherence protocol for a write-back cache. Suppose three processors P1, P2, and P3 are in Shared state for a data block X. Processor P1 writes X at time T1. Then, Processor P2 reads X at time T2. Afterwards, Processor P3 writes X at time T3. Also, T1, T2, and T3 are sufficiently far apart. Please write down and explain all the state transitions, bus activities and actions of P1, P2 and P3 at time T1, T2, and T3.

2. (15%) One of the methods to reduce miss rate is pre-fetching. Assume it takes 2 extra cycles if the instruction misses in the cache is found in the Pre-fetch buffer. Suppose that the pre-fetch hit rate is 20% and instruction cache miss rate 2%. Hit time takes 2 cycles, and miss penalty takes 100 cycles. What is the effective miss rate using instruction pre-fetching?

3. (15%) For a RAID 6 system illustrated in the figure below. If Disk 1 and the Row Parity disk both fail at the same time, what would be the recovery order of all the data blocks?



4. (10%) For virtual memory
   (a) Does it use set associative, fully associative or direct map scheme? Why?
   (b) What is the write strategy? Write back or write through? Why?

5. (10%) What are the advantages of PC relative conditional branch?

6. (15%) Suppose the length of the address is 32 bits. For a two-way set associative 512KB cache whose block is 128 Bytes, how many bits are used for index? How many bits are used for tag?

7. (10%) For cache coherence solutions, is a Snooping protocol or a Directory-Based Scheme more suitable for large scaled machines with large processor counts? Why?

8. (10%) Can non-blocking cache reduce miss rate? Why or why not?